SUPPLEMENTARY MATERIAL

Algorithm S1 AC-DANet Training Procedure

Require: Training dataset $\mathcal{D} = \{(\mathbf{X}_i, y_{\text{main},i})\}_{i=1}^{N}$, EEG data $\mathbf{X} \in \mathbb{R}^{C \times T}$
Require: Hyperparameters: learning rate η, epochs E, batch size B
Require: Loss weights: λ_{aux}, λ_{L1}
Ensure: Trained AC-DANet model parameters Θ

Initialize AC-DANet model parameters Θ.
Apply prior-based initialization to the `Source_Projection` weights $\mathbf{W}_{source}$.
Initialize optimizer (e.g., Adam) with learning rate η.
for epoch = 1 to E **do**
  for each batch $(\mathbf{X}, y_{\text{main}})$ in $\mathcal{D}$ **do**
    // 1. Forward Pass
    $(\mathbf{o}_{\text{main}}, \mathbf{o}_{\text{aux}}) \leftarrow$ AC_DANet_Forward$(\mathbf{X}, \Theta)$ ▷ *See Algorithm 2 for details*
    // 2. Loss Calculation
    $y_{\text{aux}} \leftarrow$ is_target(y_{main}) ▷ *Create binary auxiliary labels*
    $\mathcal{L}_{\text{main}} \leftarrow$ CrossEntropy$(\mathbf{o}_{\text{main}}, y_{\text{main}})$ ▷ *Primary task loss*
    $\mathcal{L}_{\text{aux}} \leftarrow$ CrossEntropy$(\mathbf{o}_{\text{aux}}, y_{\text{aux}})$ ▷ *PTL auxiliary loss*
    $\mathcal{L}_{L1} \leftarrow ||\mathbf{W}_{source}||_1$ ▷ *L1 sparsity loss for source filters*

    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{main}} + \lambda_{aux} \cdot \mathcal{L}_{\text{aux}} + \lambda_{L1} \cdot \mathcal{L}_{L1}$
    // 3. Backpropagation and Optimization
    $\nabla_\Theta \mathcal{L}_{\text{total}} \leftarrow$ Backpropagate$(\mathcal{L}_{\text{total}})$
    $\Theta \leftarrow$ OptimizerStep$(\Theta, \nabla_\Theta \mathcal{L}_{\text{total}})$
  end for
end for
return Θ

Algorithm S2 AC-DANet Forward Pass

Function AC_DANet_Forward$(\mathbf{X}, \Theta)$
// 1. Multi-modal Neural Representation Embedding
$\mathbf{X}_{Temp} \leftarrow \mathbf{X}$ ▷ *Temporal Stream*
$\mathbf{X}_{ERSP} \leftarrow |\text{STFT}(\mathbf{X})|$ ▷ *Spectro-temporal Stream*
$\mathbf{X}_{Source} \leftarrow$ Source_Projection$(\mathbf{X})$ ▷ *Constrained Source Stream*

// 2. Ventral Attention Pathway (Bottom-up Saliency)
$\mathbf{H}_{VAN} \leftarrow \text{TFDC}(\mathbf{X}_{ERSP})$ ▷ *Time-Frequency Decoupled Convolutions*
$\mathbf{Z}_{VAN} \leftarrow \text{Project}(\text{Pool}(\mathbf{H}_{VAN}))$ ▷ *Generate Key & Value sequence*

// 3. Dorsal Attention Pathway (Top-down Control)
$\mathbf{X}_{DAN_in} \leftarrow \text{Concatenate}(\mathbf{X}_{Temp}, \mathbf{X}_{Source})$
$\mathbf{H}_{DAN} \leftarrow \text{TransformerEncoder}(\text{Project}(\mathbf{X}_{DAN_in}))$
$\mathbf{q}_{DAN} \leftarrow \text{AveragePool}(\mathbf{H}_{DAN})$ ▷ *Generate Query vector*

// 4. Gated Cross-Attention Fusion (GCAF)
$\mathbf{z}_{attn} \leftarrow \text{CrossAttention}(\text{Query} = \mathbf{q}_{DAN}, \text{Key/Value} = \mathbf{Z}_{VAN})$
$\mathbf{g} \leftarrow \sigma(\mathbf{W}_g \mathbf{q}_{DAN} + \mathbf{b}_g)$ ▷ *Compute modulatory gate*
$\mathbf{z}_{fused} \leftarrow \mathbf{g} \odot \mathbf{z}_{attn}$ ▷ *Apply gate to attended features*

// 5. Parietal Target Locus (PTL) Pathway
$\mathbf{x}_{Pz} \leftarrow$ Select_Pz_Channel$(\mathbf{X})$
$(\mathbf{s}_{conf}, \mathbf{o}_{aux}) \leftarrow \text{PTL_CNN}(\mathbf{x}_{Pz})$ ▷ *Get score and auxiliary logits*

// 6. Final Classification
$\mathbf{z}_{final} \leftarrow \text{Concatenate}(\mathbf{z}_{fused}, \mathbf{s}_{conf})$
$\mathbf{o}_{\text{main}} \leftarrow$ MLP_Classifier$(\mathbf{z}_{final})$

return $(\mathbf{o}_{\text{main}}, \mathbf{o}_{aux})$
End Function

TABLE SI

BEHAVIORAL DATA OF SOUND TARGET DETECTION FOR ALL SUBJECTS UNDER DIFFERENT SNR

	High SNR		Low SNR	
	Success Rate	$\overline{RT}$ (ms)	Success Rate	$\overline{RT}$ (ms)
S1	100%	465.16	90.67%	535.59
S2	97.33%	578.7	93.33%	630.17
S3	92.00%	599.63	76.00%	691.14
S4	100%	528.57	80%	631.97
S5	97.33%	616.89	82.67%	663.46
S6	98.67%	497.25	88%	600.41
S7	96%	492.3	88%	546.65
S8	92%	508.63	88%	569.67
S9	100	467.45	90.67	536.8
S10	96	494.32	88	544.42
S11	98.67	497.42	88	601.37
S12	100	531.5	80	629.48
Mean	97.33	523.15	86.11	598.43
Std±	2.78	47.97	5	50.09

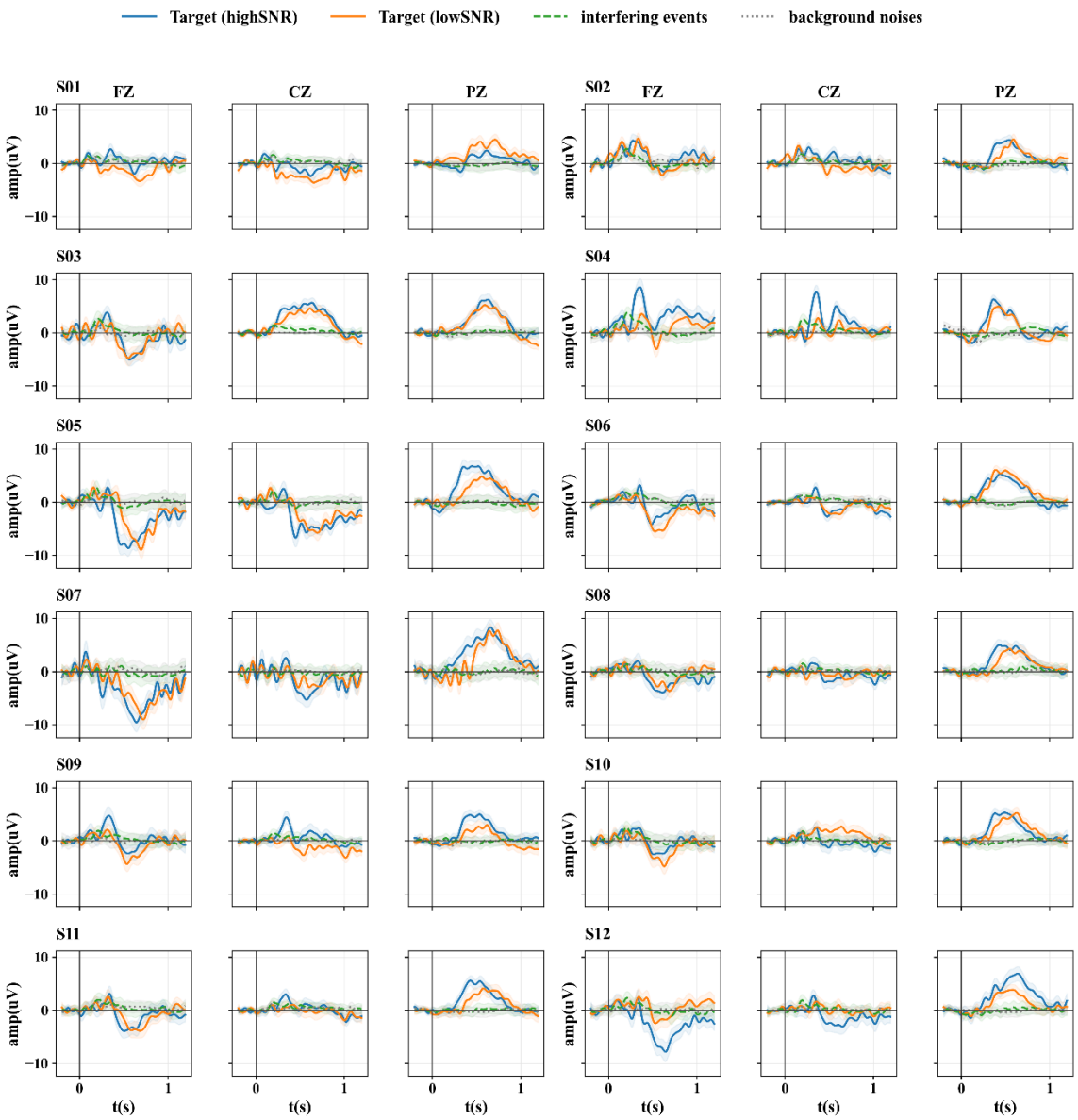

Fig. S1. Multi-channel ERP curves for each participant.

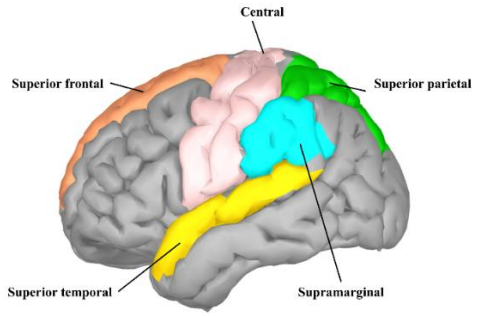

Fig. S2. The brain regions and locations of interest in this paper.

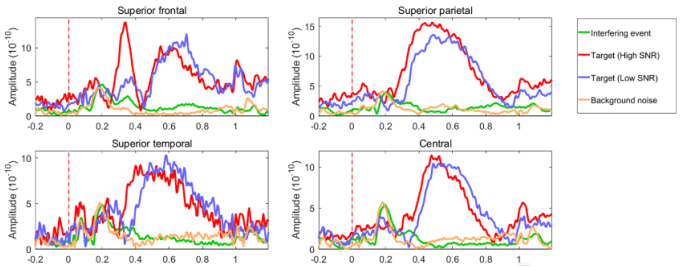

Fig. S3. Activation curves of the superior frontal gyrus, superior temporal gyrus, central area and parietal gyrus under four conditions.

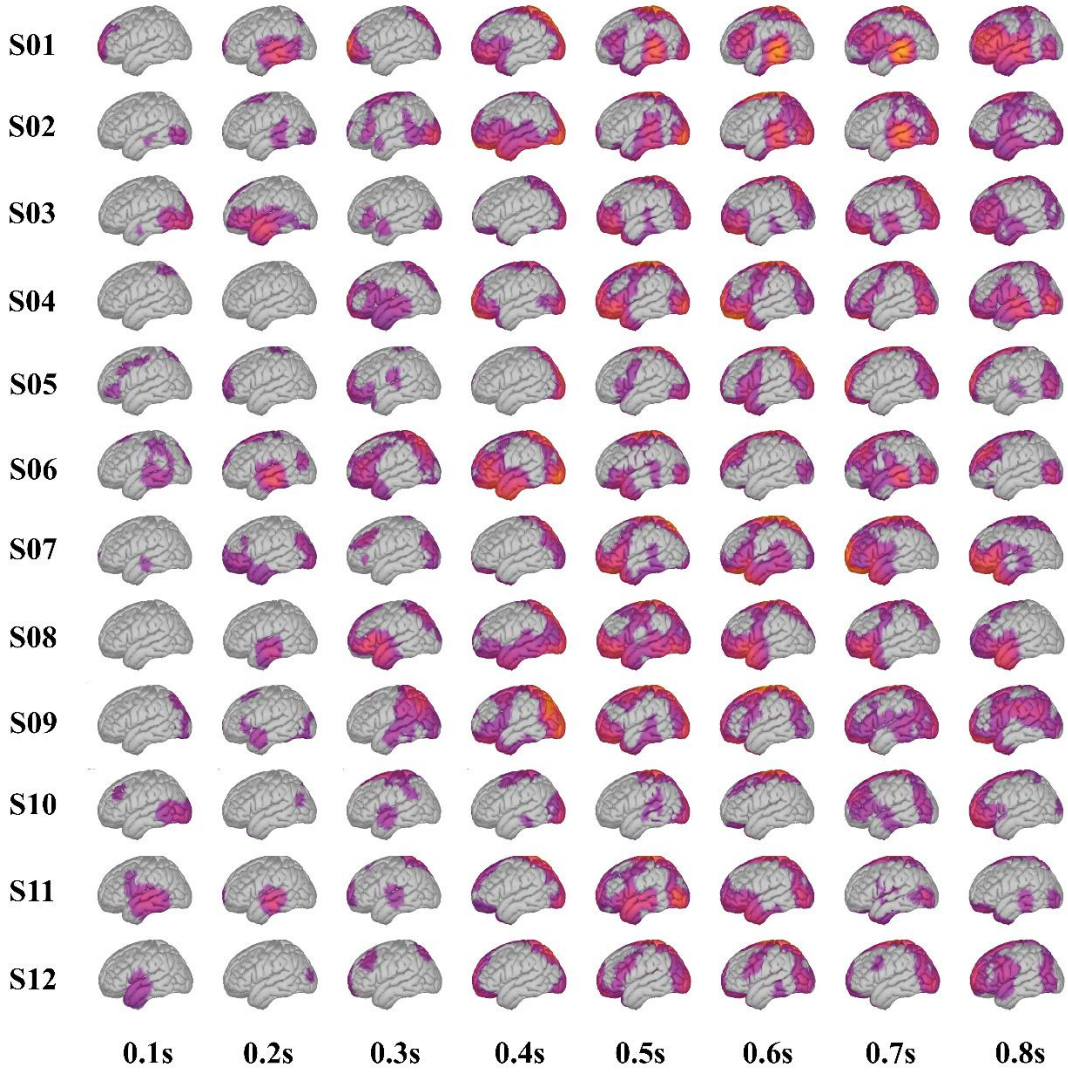

Fig. S4. Source space activation for each participant (target samples).

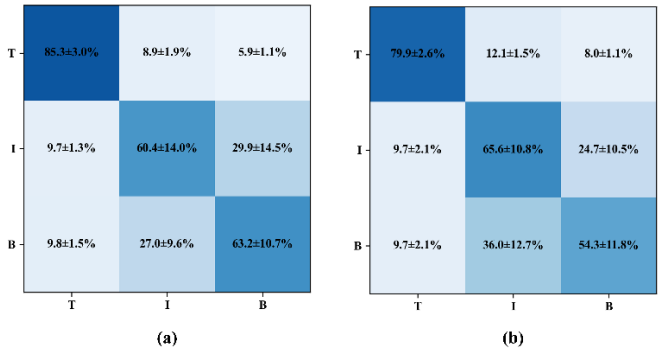

Fig. S5. The confusion matrix for the three-class classification results within-subject (a) and cross-subject (b).

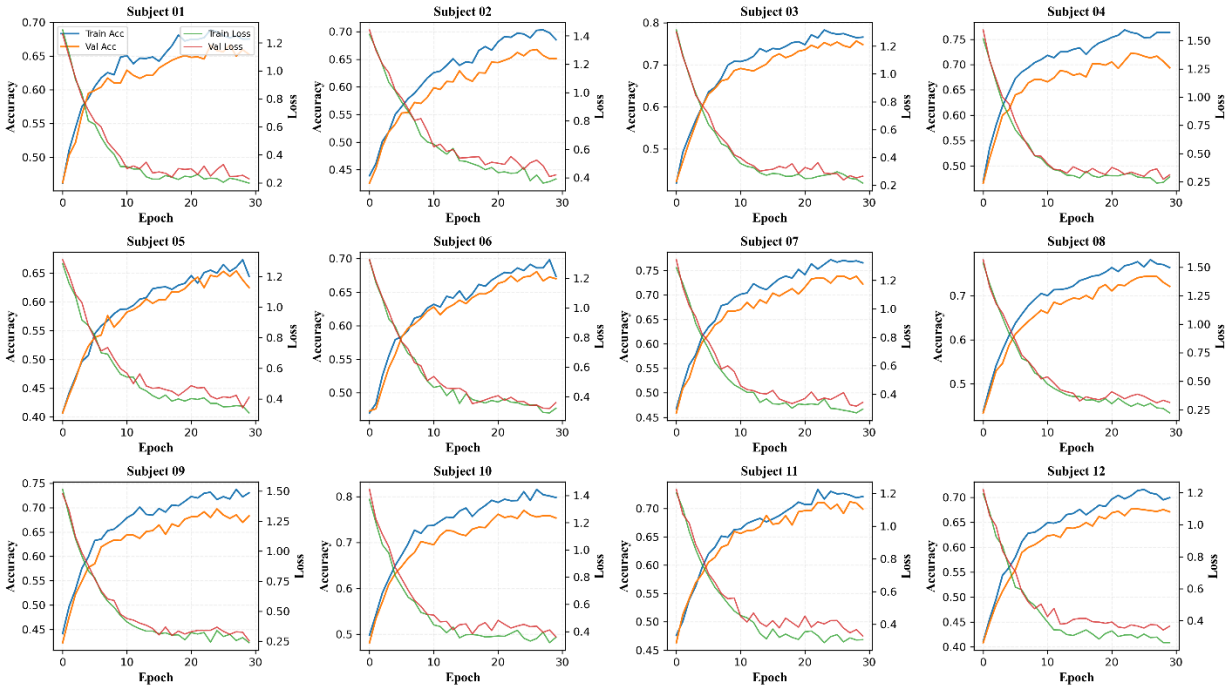

Fig. S6. Within-subject training convergence curves (loss and accuracy).

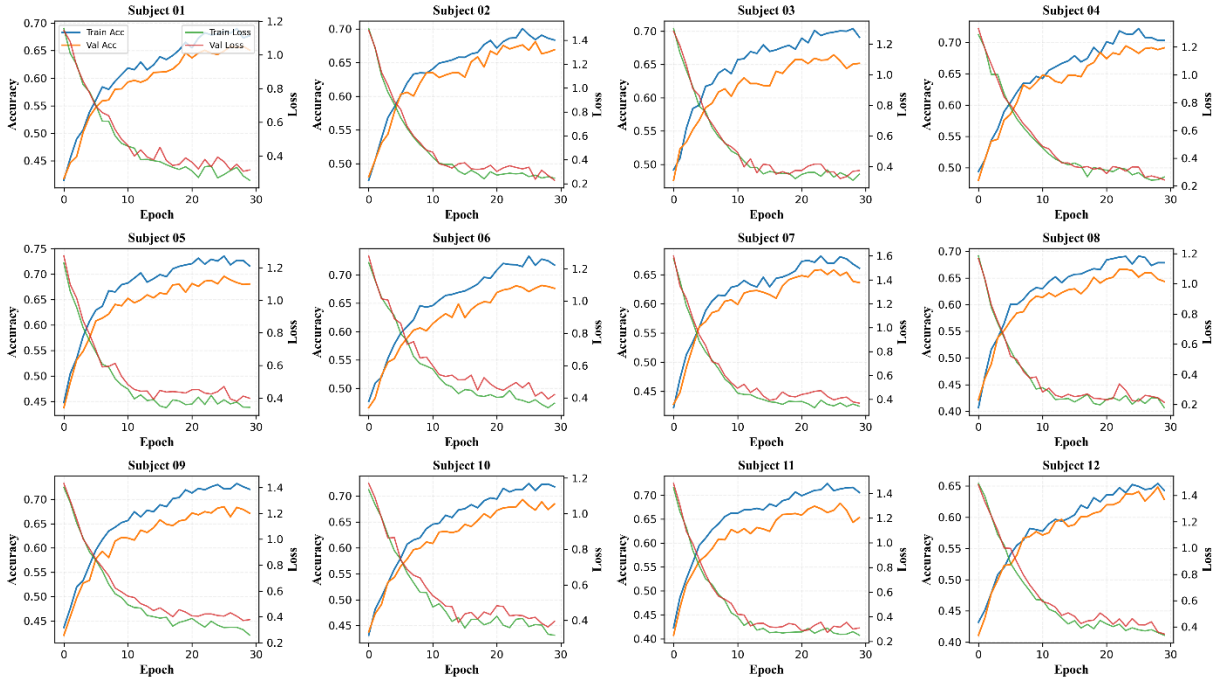

Fig. S7. Cross-subject training convergence curves (loss and accuracy).

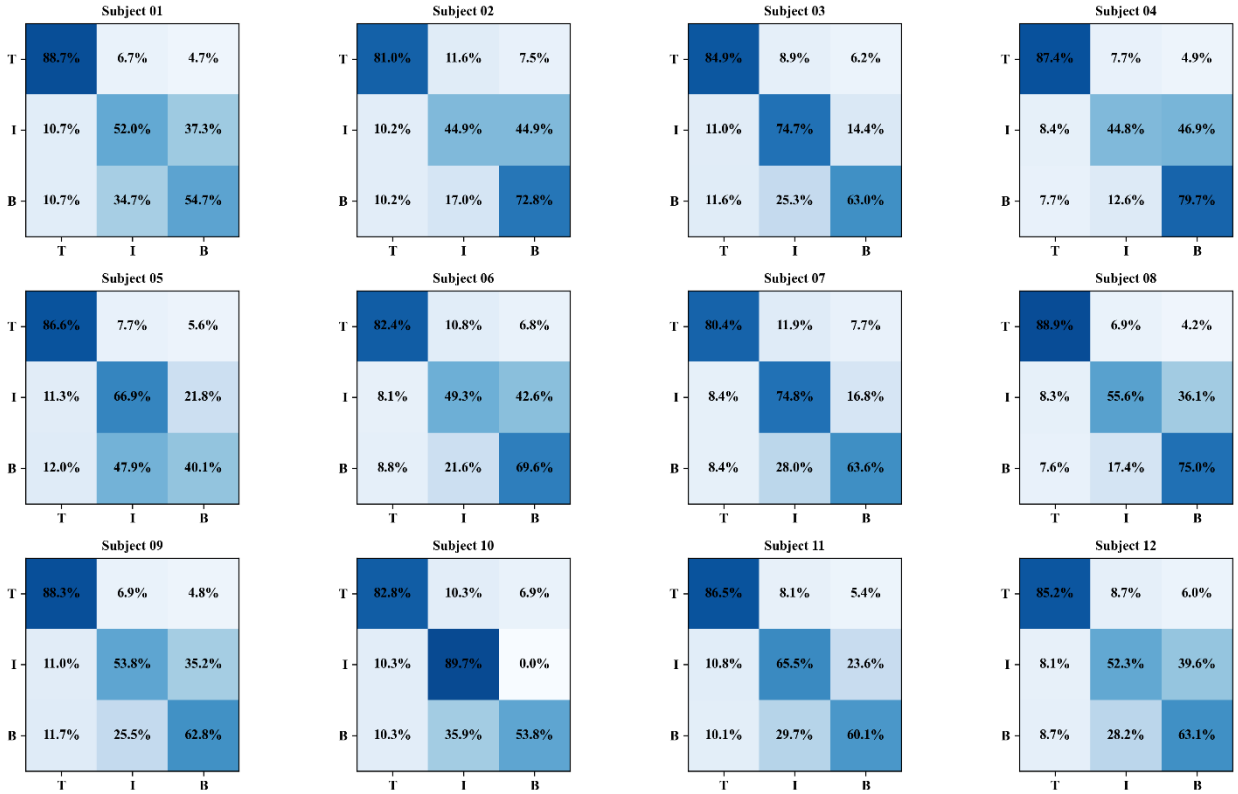

Fig. S8. Confusion matrix for each within-subject participant.

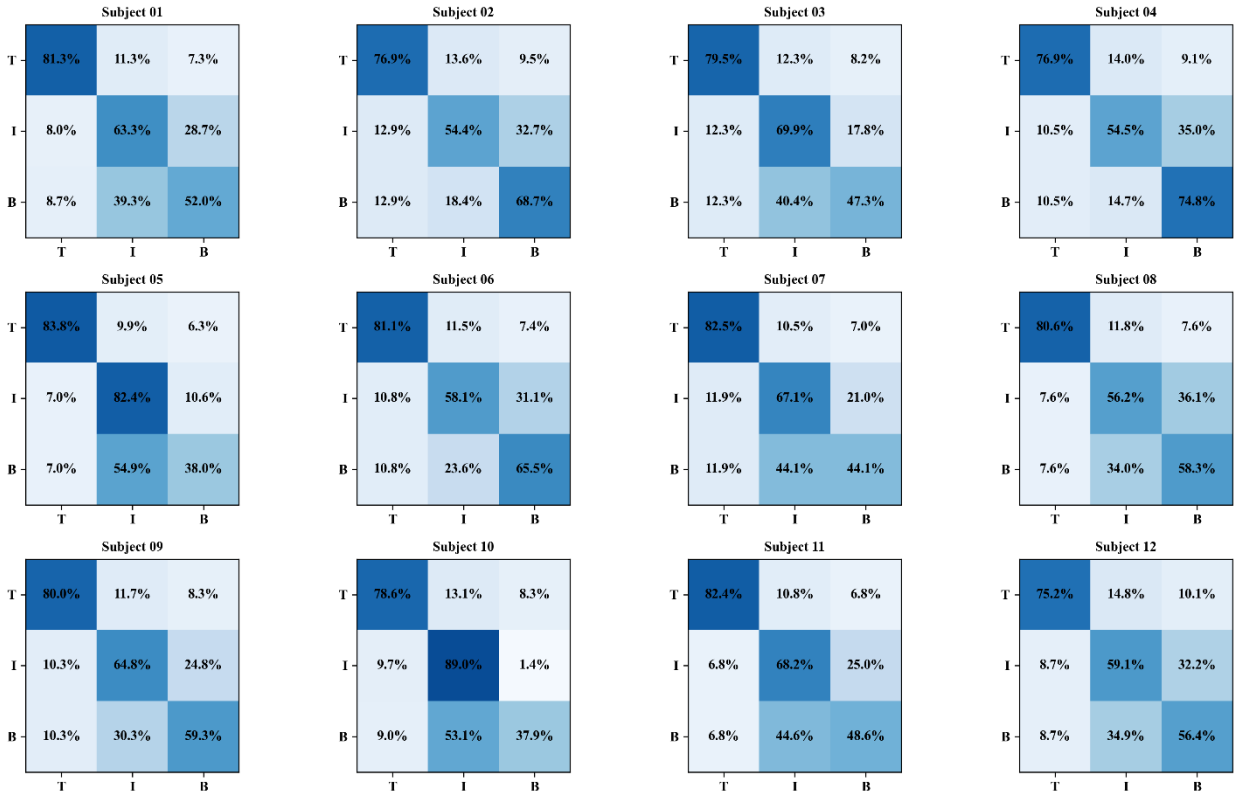

Fig. S9. Confusion matrix for each cross-subject participant.

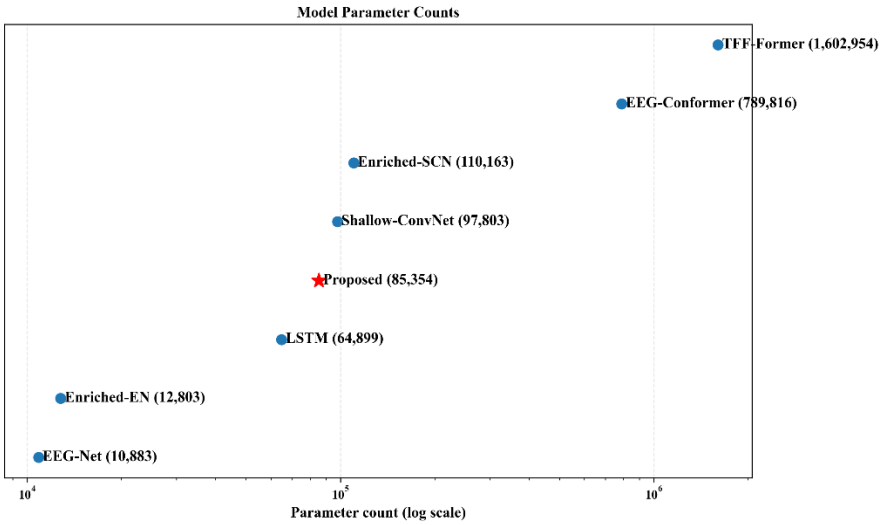

Fig. S10. Comparison of the algorithm's parameter count.

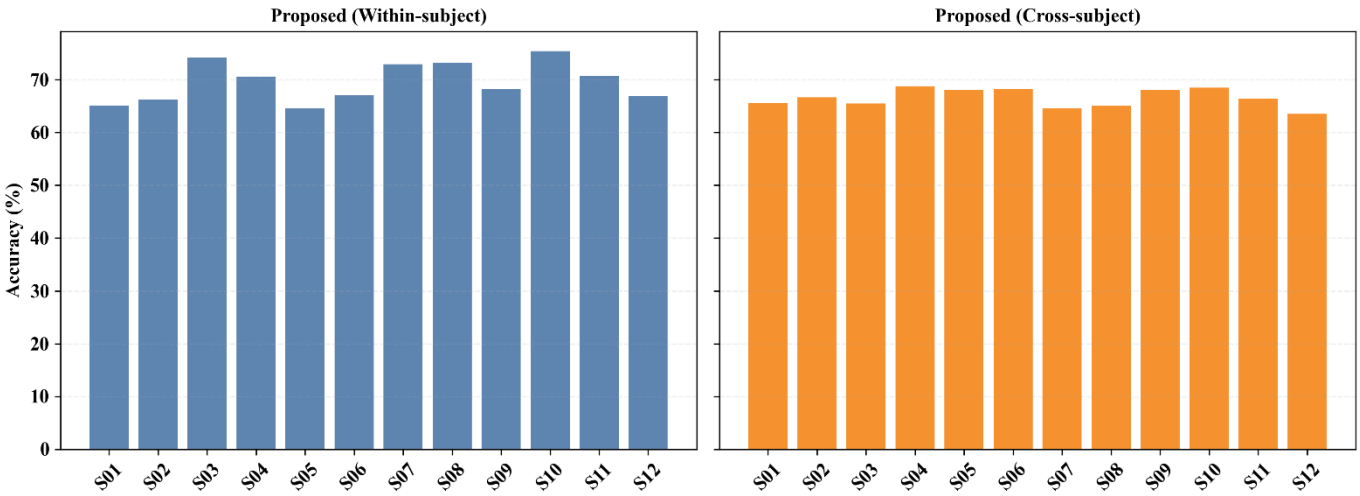

Fig. S11. Individual accuracy for all participants, within-subject (left) and cross-subject (right).